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Cost-Effective and Innovative Well Offloading and Testing: A Jet Pump Success Story from Iraq

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Abstract

As the petroleum industry navigates increasing economic pressures, operators continue to seek cost-effective production solutions that do not compromise safety or operational efficiency. One such solution is the application of jet pump technology, which offers a rigless, flexible method for well offloading and testing. This paper presents a field case from Iraq where a jet pump was successfully implemented as an alternative to traditional coiled tubing and nitrogen-based offloading, demonstrating strong performance in terms of both flow assurance and cost savings.

In 2023, a newly drilled well targeting three distinct production zones in southern Iraq was evaluated for its inflow potential. Historically, wells in this region were offloaded using coiled tubing combined with nitrogen injection, a common but costly approach that involves logistical complexities and long mobilization times (Lea et al., 2008). For this operation, the engineering team selected a hydraulic jet pump system designed using Jet Evaluation and Modeling Software (JEMS), which enabled precise nozzle and throat sizing tailored to wellbore pressure, temperature, and fluid properties (Pugh, 2009).

The jet pump was deployed via slickline, offering a low-risk, non-intrusive deployment method. High-pressure power fluid was injected down the annulus, while produced fluids were returned through the tubing. Production zone testing was executed by sequentially opening individual Sliding Sleeve Doors (SSDs), allowing controlled unloading and isolated flow assessment from each zone. The system achieved a peak production rate of approximately 2,200 barrels per day (BPD), well above initial design targets.

In terms of cost and operational impact, the jet pump system significantly outperformed conventional nitrogen lift. The use of rigless installation and surface-based parameter tuning eliminated the need for nitrogen tankers and reduced rig time by seven days, contributing to an overall cost savings of over 500% when compared to previous nitrogen-based operations in the same field. The field team would also be able to change nozzle and throat configurations on-site, further optimizing drawdown and pump efficiency. This level of flexibility is a well-documented advantage of jet pump systems, especially in early production scenarios or wells with high water cut and unstable flow regimes (Takács, 2015; Brown, 1977).

This study reinforces the economic and operational benefits of jet pumps for initial well unloading and production evaluation. Their ability to be rapidly deployed, adjusted, and retrieved without a rig gives them an edge over conventional artificial lift techniques during early well life. In replacing nitrogen offloading, jet

pumps not only reduce cost but also improve control over flowback and data acquisition. The findings from this case provide a practical and scalable model for other operators seeking to transition from conventional nitrogen kick-off to jet pump-based well stimulation strategies.

Artificial Lift Systems Overview

Artificial lift systems (ALS) are essential for enhancing well productivity, especially as natural reservoir energy declines over time. The selection of the most appropriate lift method depends on several variables, including reservoir pressure, fluid properties, well trajectory, sand content, flow rate targets, and economic constraints. Among the most widely adopted ALS technologies are Electrical Submersible Pumps (ESP), Gas Lift, Hydraulic Jet Pumps, Progressing Cavity Pumps (PCP), and Reciprocating Rod Lift (RRL); each offering unique advantages and operational limitations depending on well conditions.

- Electrical Submersible Pumps (ESP) are widely used in offshore and high-volume onshore wells due to their ability to handle large fluid rates. These systems consist of a downhole centrifugal pump powered by an electric motor connected via cables along the production tubing. While highly effective in high-rate vertical wells, ESPs are sensitive to gas and solids, suffer from high failure rates in abrasive environments, and typically require costly rig-based interventions for retrieval or servicing (Brown, 1977; Takács, 2015). These characteristics make them less suitable for early-life, low-rate, or intervention-sensitive wells.
- Gas Lift systems operate by injecting high-pressure gas into the casing-tubing annulus to reduce the hydrostatic head and stimulate flow. This method is highly flexible and compatible with deviated or horizontal wellbores, and it performs well in high-GOR environments. However, it also presents challenges such as flow instabilities (slugging, heading), optimization complexity, and dependency on reliable gas supply infrastructure. In depleted or low-pressure wells, maintaining adequate injection pressure can become operationally burdensome (Lea et al., 2008).
- Hydraulic Jet Pumps offer a rigless, highly adaptable form of artificial lift that is especially suitable for early well life, high water cut, and multiphase flow conditions. These systems work by injecting a high-pressure power fluid, typically produced water or treated brine, downhole, creating a low-pressure zone that draws in reservoir fluids. The mixed stream is then lifted to the surface through the tubing. Jet pumps are slickline retrievable, have no moving parts, and are tolerant to solids and gas. Their design flexibility and surface-level controllability make them ideal for well offloading, short-term testing, and production kickoff scenarios in cost-sensitive environments (Pugh, 2009; Brown, 1977).
- Progressing Cavity Pumps (PCP) utilize a surface-driven helical rotor-stator assembly to produce fluids. They are particularly well-suited to handling viscous and sand-laden flows, and operate efficiently at low speeds, making them ideal for heavy oil or shear-sensitive emulsions. However, PCPs have operational limitations in high-temperature or chemically aggressive conditions due to elastomer degradation. Additionally, rotor-stator wear becomes a concern in highly abrasive environments or extended operations (Takács, 2015).
- Reciprocating Rod Lift (RRL), or sucker rod lift, is among the most established artificial lift methods. It uses a surface beam pump connected to a rod string that drives a plunger inside a downhole pump barrel. RRL systems are easy to maintain and cost-effective in shallow wells producing at low-to-moderate rates. However, they are poorly suited for deviated or horizontal wells due to rod buckling and mechanical losses, and have practical depth limitations (Lea et al., 2008).

In the case described in this paper, the operator selected a jet pump system to offload and test a newly drilled well in southern Iraq. This decision was based on several critical factors: the moderate production forecast, cost sensitivity, desire for a rigless solution, and the need for flexible, retrievable equipment capable of handling early-life well dynamics. Compared to conventional nitrogen kick-off or ESP-based production systems, the jet pump offered a low-risk, efficient, and adaptable solution for both offloading and early production evaluation.

Table 1 summarizes the key advantages and operational limitations of common artificial lift systems. The selection process must weigh both technical and economic factors. In the case of Well T46S, the need for a flexible, retrievable, and low-intervention method made hydraulic jet pumping the most suitable option for well offloading and initial production testing.

Table 1—Key advantages and limitations of different artificial lift systems

Artificial Lift Method	Advantages	Operational Limitations
Electrical Submersible Pump (ESP)	- High flow rate capability - Suitable for deep and vertical wells - Compact surface footprint	- Poor gas/solids handling (gas lock) - High OPEX due to rig-based retrieval - Limited flexibility to varying rates
Gas Lift	- Flexible with variable rates - Suitable for deviated/horizontal wells - Long operational life	- Flow instabilities (slugging) - Needs continuous gas supply and compressors - Reduced efficiency in deep wells
Hydraulic Jet Pump	- No moving parts - Handles gas, solids, and sand - Fully retrievable via slickline - Adaptable to well conditions	- Requires clean power fluid - Surface facilities are complex - Sensitive to drawdown and backpressure design
Progressing Cavity Pump (PCP)	- Excellent for viscous and sand-laden fluids - Low shear - Simple surface drive system	- Depth-limited (<2,500 m) - Elastomer degradation at high temperature - Torque challenges in deviated wells
Reciprocating Rod Lift (RRL)	- Simple and cost-effective for shallow wells - Easy surface monitoring - Proven reliability	- Depth and deviation limitations - Rod/tubing wear in solids - Bulky and noisy surface units

Field and Well Background

The subject of this case study is Well T46S, located in a producing oil field in southern Iraq, operated by an international oil company. The well was drilled in October 2023, targeting three hydrocarbon-bearing zones within a moderately mature reservoir. The fluid and formation characteristics include a crude oil gravity of 26.5° API, CO_2 content of 0.5%, and brine salinity of approximately 18%, indicative of moderately heavy oil and a moderate scaling or corrosion risk, typical of carbonate-dominated formations in this region.

Historically, wells in this field were offloaded using nitrogen injection via coiled tubing, a standard but capital-intensive technique for removing completion fluids and initiating flowback in low-pressure or static wells. While nitrogen kick-off can rapidly unload wells, it poses several operational drawbacks. These include high mobilization costs, dependency on specialized nitrogen pumping units, the requirement for coiled tubing, and prolonged rig time during the job. Moreover, nitrogen lift does not enable continuous flow monitoring or data acquisition during the unloading process, which limits its use for reservoir evaluation or short-term testing applications (Lea et al., 2008; Craft & Hawkins, 1991).

To address these limitations, the operator selected a hydraulic jet pump system for offloading and early production testing of Well T46S, marking the first application of this technology in the field. The pump was deployed and seated at a depth of 1,014 meters, using a 2.56-inch Sliding Sleeve Door (SSD) integrated into the completion design. This depth was selected to allow sufficient drawdown across all three zones while maintaining a safe standoff from pump cavitation thresholds and avoiding excessive backpressure on the formation.

Jet pump technology was chosen for its retrievability, low operational risk, and its ability to function efficiently in high water cut and multiphase flow conditions (Pugh, 2009). Additionally, jet pumps allow surface-based tuning and real-time performance monitoring, enabling the operating team to optimize drawdown and power fluid conditions based on live production data. This contrasts with nitrogen injection, which provides no control once initiated and requires significant surface logistics.

This paper presents the end-to-end workflow for the jet pump implementation, including system design, downhole configuration, surface equipment layout, operational execution, and performance monitoring. The results demonstrate the technical feasibility and cost-efficiency of jet pumps for well unloading in this region and provide a replicable framework for future applications where conventional nitrogen lift may be operationally or economically constrained.

Jet Pump Design and Optimization

Hydraulic jet pumping was selected for Well T46S as a cost-effective, rigless, and low-intervention solution for offloading and initiating early production. The primary design objective was to successfully unload the well and achieve a target production rate of approximately 1,000 barrels per day (BPD), while maintaining flexibility in system control and minimizing the need for downhole intervention. Jet pumps are particularly suited to this role due to their retrievability, resilience to solids and gas, and adaptability to changing fluid conditions (Pugh, 2009; Takács, 2015).

Initial production conditions indicated a high water cut of approximately 80%, typical of early flowback following drilling and completion operations. The produced fluids were multiphase, with potential for varying flow regimes and gas liberation. Given these uncertainties, produced water was selected as the initial power fluid, as it minimized fluid handling requirements and enabled a closed-loop circulation system between the vessel cleaning unit and surface injection pump. This choice also reduced the cost and complexity associated with sourcing clean water or external brine, and supported sustainability goals by reusing formation water (Brown, 1977).

Design modeling was carried out using Jet Evaluation and Modeling Software (JEMS), which enabled sensitivity analysis on nozzle/throat selection, pump placement, and injection pressure to achieve the target drawdown. The design considered bottomhole pressure, reservoir fluid properties, and flow assurance constraints to prevent cavitation or backpressure limitations. Jet pump efficiency was optimized by selecting an 11C nozzle/throat combination in a reverse circulation jet pump, suited to the high liquid volume fraction and expected pump intake pressure, aligning with jet pump performance criteria (Pugh, 2009).

The final jet pump system configuration was set to operate under these baseline parameters, summarized in Table 2, with the ability to be tuned at surface as production stabilized. No downhole modifications or retrievals were anticipated, and all system adjustments could be made in real-time via surface controls.

Table 2—Key parameters for jet pump system installation

Parameter	Value
Jet Pump Model	OilMaster – Reverse Circulation
Nozzle/Throat Combination	11C N/T
Power Fluid Type (initial)	Produced Water
Injection Rate	1,850 BPD
Injection Pressure	2,000 psi
Production Target	1,000 BPD
Water Cut (Initial)	80%
Pump Setting Depth	1,014 meters
Jet Pump Location	2.56” Sliding sleeve door (SSD)
Design Software Used	JEMS (Jet Pump Efficiency Modeling System)

Design Workflow and Considerations

The hydraulic jet pump was selected for Well T46S based on its proven ability to operate under dynamic fluid conditions and accommodate wide variations in production rate and drawdown pressure. Its lack of moving parts and surface-level controllability make it particularly suitable for early well life when flow conditions may shift significantly over short durations (Pugh, 2009). A reverse circulation jet pump was chosen and installed at a depth of 1,014 meters, precisely seated on a 2.56" Sliding Sleeve Door (SSD). This configuration allowed for rigless deployment via slickline and retrieval without workover, preserving operational flexibility and reducing mobilization costs (Brown, 1977).

During the initial flowback phase, produced water was used as the power fluid at high injection pressure to ensure sufficient suction energy and early well unloading. As the well transitioned to a more stable production regime and the water cut dropped from ~80% to ~13%, the system smoothly handled oil-dominant flow conditions without requiring any hardware change or retrieval, highlighting the adaptability of the design (Takács, 2015).

This 11C nozzle throat combination was optimized to deliver the desired production rate, while maintaining manageable backpressure and maximizing pump efficiency across the expected operating envelope. The decision to use reverse circulation was based on its superior tolerance to solids and fines, often encountered during initial production or in unconsolidated formations. This approach also helped mitigate potential erosion and scale deposition near the jet pump throat area, extending tool life and maintaining performance (Pugh, 2009; Lea et al., 2008).

Optimization Strategy

Jet pump performance for Well T46S was continuously monitored and optimized using a combination of surface and downhole pressure readings, real-time fluid rate measurements, and periodic water cut analysis. As production stabilized, the power fluid injection rate was incrementally adjusted to reduce energy consumption while sustaining the necessary drawdown. This operational tuning was executed entirely from the surface, with no downhole intervention required, highlighting one of the major advantages of jet pump systems, their surface-level modularity and responsiveness (Kalwar et al., 2022; Pugh, 2009).

The system's flexibility was particularly valuable as the well transitioned from water-dominant to oil-dominant production, with the water cut decreasing from approximately 80% to 13% during flowback. This transition occurred without any loss of pump efficiency or need for mechanical reconfiguration, consistent with findings from other SPE-documented case studies of jet pump systems in early production environments (Chavan et al., 2012; Shen, 2010).

To validate the pump design and confirm drawdown targets, an Inflow Performance Relationship (IPR) analysis was carried out during the planning phase. The IPR curve (Fig. 1 below) provided critical insight into the well's production potential under variable bottomhole flowing pressure (BHFP) conditions. It also served as a decision-making tool for selecting the appropriate nozzle/throat combination, drawdown range, and expected production rate envelope.

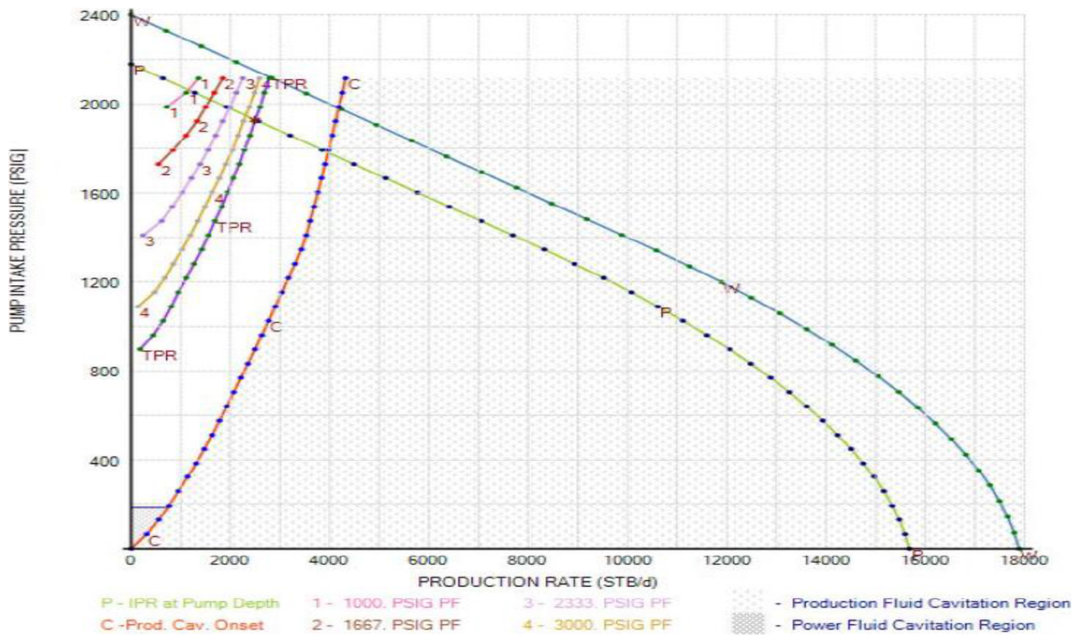

Figure 1—IPR Curve with Jet Pump Operating Point

For this analysis, Vogel's equation was used to model the inflow profile under solution-gas drive conditions, accounting for multiphase behavior and anticipated formation pressure depletion. The results indicated a moderate inflow potential, consistent with known characteristics of the field and expected reservoir energy after drilling. This modeling helped ensure that the jet pump would operate within an efficient range throughout both the unloading and early production periods (Vogel, 1968; Hatzlavramidis, 1991).

By integrating IPR analysis into the jet pump design workflow, the engineering team was able to optimize pump placement, reduce injection overpressure, and ensure alignment between reservoir behavior and lift system performance, maximizing production efficiency without risking formation damage or pump cavitation.

Field Execution and Monitoring

Following completion of the design and simulation phase, the hydraulic jet pump system was mobilized to Well T46S in southern Iraq. The rig-up process, comprising surface pump installation, power fluid circulation line connections, and pressure testing, was completed efficiently within a single working day. A reverse circulation jet pump with an 11C nozzle/throat configuration was selected and deployed via slickline to a setting depth of 1,014 meters, aligned with the 2.56" side-pocket SSD included in the completion. This rigless, intervention-free approach eliminated the need for coiled tubing or a workover rig, significantly reducing mobilization time and operational cost (Chavan et al., 2012; Kalwar et al., 2022).

Once surface integrity tests were confirmed, power fluid injection was initiated using produced water, circulated at 1,850 BPD and 2,000 psi through surface triplex pumps. Within minutes of startup, the jet pump established sufficient drawdown to induce inflow from the reservoir. The well was fully offloaded within 12 hours, and stable production was achieved, allowing immediate transition into the extended production testing phase. This rapid and controlled unloading is consistent with prior SPE-documented jet pump field cases, where early production and flowback are enabled without nitrogen logistics or large crews (Shen, 2010; Clark et al., 2014).

Throughout the operation, fluid samples were collected at regular intervals to assess water cut behavior, monitor solid content, and evaluate the reservoir's pressure response. As production progressed, the water cut declined from ~80% to ~13%, confirming effective drawdown and cleanup. Importantly, no retrieval of the jet pump was required during the test. All performance adjustments, including changes in injection pressure and surface pump rate, were carried out at surface, underscoring the operational flexibility and real-time responsiveness of the jet pump system (Pugh, 2009).

The field results validated the pre-job modeling and reinforced the jet pump's role as a rigless, data-driven solution for early production testing. The smooth deployment, fast offloading of the well, and stable flow profile demonstrated the system's technical robustness and economic viability for similar wells under comparable reservoir and logistical constraints.

Production Performance and Optimization

The jet pump system delivered strong performance throughout the test phase of Well T46S, validating its selection for early production and well offloading. The well produced at a sustained rate of 2,200 BPD, significantly surpassing the initial design forecast of 1,000 BPD. Simultaneously, the water cut declined from 80% to 13%, indicating effective cleanup of completion fluids and smooth transition into reservoir-driven stabilized flow. Throughout the operation, no cavitation, mechanical wear, or solids-related complications were observed, underscoring the durability and tolerance of the reverse circulation jet pump design (Pugh, 2009; Kalwar et al., 2022).

All optimization efforts, including flow rate control and pressure tuning, were conducted at the surface, eliminating the need for downhole intervention or mechanical retrieval. This allowed the production team to dynamically respond to real-time conditions without disrupting flow or increasing operational risk. The system demonstrated high resilience to changing fluid properties, a behavior consistent with findings from previous SPE case studies of jet pump applications in early production and brownfield environments (Chavan et al., 2012; Mei et al., 2017).

The execution phase of the project yielded several technical and operational benefits, reinforcing the suitability of jet pump systems for similar applications:

- The mobilization and rig-up of surface equipment and subsurface deployment were completed within one day, requiring no rig mobilization or heavy equipment; a key advantage in remote or infrastructure-constrained fields.
- Flow initiation occurred immediately upon startup of the power fluid system. This rapid response eliminated the need for extended well stimulation, commonly required in nitrogen-based unloading scenarios (Shen, 2010).
- The well was fully offloaded within 12 hours, establishing stable flow and allowing seamless transition to production testing without delays or surface instability.
- The system achieved more than double the expected production rate, with a concurrent and substantial decrease in water cut, validating the drawdown strategy and reservoir response modeling.
- Crucially, all tuning and optimization were executed via surface controls; no downhole mechanical adjustments were needed, and slickline retrieval was never required during the entire test phase.
- As a result, the system minimized downtime, eliminated non-productive rig time, and enabled a safe, low-exposure operational workflow aligned with field HSE requirements (Clark et al., 2014).

These outcomes confirm the technical feasibility, economic advantage, and operational flexibility of using hydraulic jet pumps for early production applications. The ability to initiate, monitor, and optimize well drawdown with a high level of control and without intervention provides a compelling alternative to nitrogen lift and other conventional unloading methods.

Operational and Running Procedure

The successful application of the hydraulic jet pump in Well T46S was enabled by a clearly defined operational strategy and close collaboration between the field operations and engineering teams. The planning phase began with a detailed evaluation of reservoir pressure, expected fluid composition, and wellbore geometry. In conjunction with jet pump specialists, the operator conducted a comprehensive pre-installation review to align downhole objectives with surface capabilities.

Simulation studies were performed using Jet Evaluation and Modeling Software (JEMS), allowing the team to model expected drawdown, pump efficiency curves, and power fluid requirements. These simulations guided the selection of the 11C nozzle/throat combination, and ensured the system would deliver the target production rate with 1,850 BPD injection at 2,000 psi. This level of planning is consistent with best practices in jet pump modeling and deployment, as documented in SPE field studies (Pugh, 2009; Kalwar et al., 2022).

For the surface layout, a closed-loop power fluid circulation system was implemented using produced water as the initial injection medium. Surface equipment included a triplex high-pressure pump rated to 5,000 psi, a vessel cleaning unit for solids removal, a safety control panel, and digital instrumentation for monitoring pressure and flow rates. The production line was routed through a three-phase test separator, enabling accurate fluid sampling and rate measurement during the test phase. Prior to starting operations, the full system was hydraulically pressure-tested to confirm mechanical integrity and safety compliance, minimizing operational risk during startup (Clark et al., 2014; Mei et al., 2017).

The downhole assembly was redressed at the surface, function-tested, and configured for reverse circulation, chosen for its improved solids handling and erosion resistance. The reverse circulation pump (Fig. 2) was adjusted to fit the 2.56" Sliding Sleeve Door (SSD) integrated within the well's completion string. The SSD was mechanically opened using a B-shifting tool, allowing the power fluid to enter the pump path. Deployment was carried out using slickline, and the pump was successfully seated at a measured depth of 1,014 meters. The operation required no rig or coiled tubing, significantly reducing field cost, complexity, and HSE exposure, a major advantage over traditional well stimulation workflows (Chavan et al., 2012; Shen, 2010).

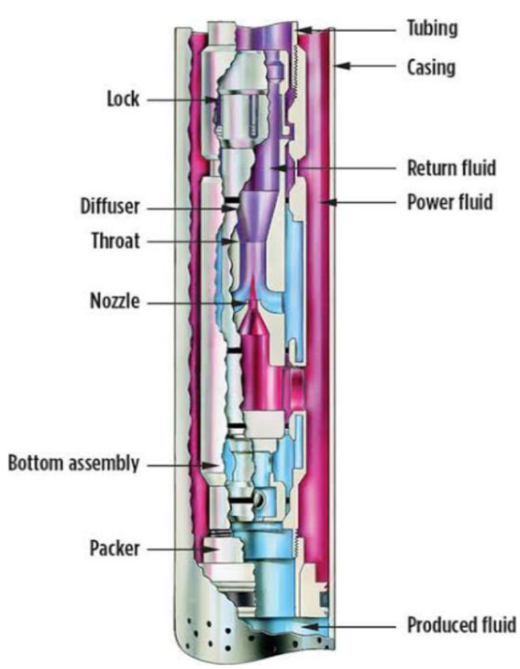

Figure 2—Reverse Circulation jet pump in SSD installation

This rigless deployment model not only accelerated execution but also enhanced adaptability, allowing the pump to be retrieved or replaced without heavy equipment if needed, highlighting the operational value of jet pump systems in cost-sensitive and infrastructure-limited field environments.

The startup phase of the jet pump system in Well T46S was smooth and immediate. Once power fluid injection commenced from the casing annulus, flowback was observed from the tubing within minutes, and the well began unloading without any delay. The offloading process was completed within 12 hours, establishing a steady-state production profile. Fluids returning to the surface were directed through a three-phase test separator, where samples were collected at regular intervals to monitor water cut, salinity, and overall fluid composition. No cavitation, flow instability, or solids-related issues were encountered during this phase, highlighting the system's mechanical reliability and its ability to tolerate minor fines or debris typically present during early production (Kalwar et al., 2022; Mei et al., 2017).

Following offloading, the operation transitioned directly into the production testing phase, maintaining the same downhole configuration. No retrieval or mechanical reconfiguration of the jet pump was necessary. All operational adjustments, such as injection pressure, surface pump rate, and wellhead choke settings, were managed entirely at surface. This rigless optimization workflow allowed for seamless testing and maximum uptime (Chavan et al., 2012).

During the test period, the production rate reached 2,200 BPD, significantly exceeding the design objective of 1,000 BPD. Concurrently, the water cut declined from 80% to 13%, reflecting both effective well unloading and successful reservoir cleanup. The jet pump sustained drawdown and responded well to changes in fluid composition, including the shift from water-dominant to oil-dominant flow, without requiring any component replacement or pull-out-of-hole (Pugh, 2009).

At the conclusion of testing, the jet pump system was shut down, and the surface layout, including power fluid lines, return manifolds, and separator connections, was depressurized and flushed in accordance with field safety protocols. The pump itself was left in place, with the option of retrieval via slickline available depending on future reservoir management plans. All operational data, including injection rates, surface and bottomhole pressures, fluid samples, and jet pump performance metrics, were compiled for post-job analysis and integrated into the field's long-term artificial lift strategy.

The operation also yielded several important technical learnings and best practices for future jet pump deployments:

- The use of clean, filtered power fluid at startup was essential in preventing nozzle/throat plugging, especially in high-solids wells.
- Gradual and incremental adjustments to injection pressure helped prevent pressure surges that could destabilize flow or cause rapid load changes on surface equipment.
- Real-time surface monitoring of injection and return lines, using digital pressure and rate sensors, enabled continuous validation of pump behavior and early detection of any anomalies.
- The ability to operate and optimize the jet pump entirely from surface, without any requirement for mechanical retrieval, significantly reduced operational time, personnel exposure, and total project cost (Clark et al., 2014; Mei et al., 2017).
- This execution further reinforces the role of hydraulic jet pumps as a reliable and cost-effective tool for early production testing, particularly in fields where intervention logistics are constrained, or operational agility is required.

Fig. 3 below represents the typical process flow diagram of a jet pump installation from planning till execution of the job.

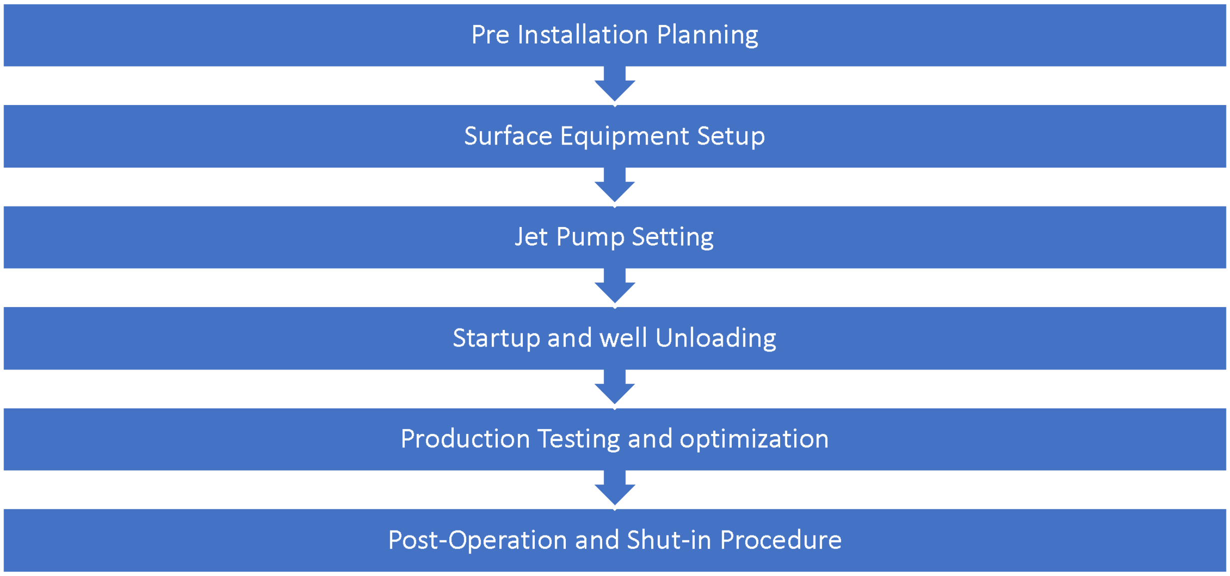

Figure 3—Operation Process Flow Diagram.

Results and Discussion

The application of a hydraulic jet pump in Well T46S achieved clearly measurable outcomes in both operational efficiency and production performance. The primary objective, to offload the well and evaluate its production capacity, was fulfilled rapidly, reliably, and without intervention. From initial rig-up to final data capture, the system delivered on key performance indicators while minimizing non-productive time, logistics burden, and HSE exposure.

Upon startup, the jet pump immediately initiated inflow, establishing a stable production profile without delay. The well was fully offloaded within 12 hours, and the operation transitioned seamlessly into the production testing phase. Throughout testing, the well consistently produced at 2,200 barrels per day (BPD), more than double the initial expectation of 1,000 BPD. Simultaneously, the water cut declined from 80% to 13%, reflecting effective drawdown and fluid cleanup from the reservoir. These results demonstrated not only the technical capability of the jet pump system but also the validity of pre-job modeling performed using Jet Evaluation and Modeling Software (JEMS), which helped optimize nozzle/throat configuration and injection parameters (Pugh, 2009; Kalwar et al., 2022).

Importantly, the system remained in place throughout the test without requiring retrieval or mechanical modification. All performance tuning was carried out at the surface, using pressure and rate adjustments—a key advantage in early production scenarios where operational flexibility and rapid response to reservoir behavior are critical. This field trial reinforced previously documented advantages of jet pump technology: rigless deployment, high adaptability, tolerance to variable flow regimes, and surface-based control (Clark et al., 2014; Chavan et al., 2012).

In conclusion, the successful deployment in Well T46S demonstrates that hydraulic jet pump systems offer a practical, cost-effective, and reliable alternative to conventional well offloading techniques such as nitrogen lift or coiled tubing stimulation, see Table 3 for comparison in this case. The technology's performance in this case supports its broader application in early production testing, marginal well development, and intervention-sensitive operations, particularly in environments where time, cost, and safety remain top priorities.

Table 3—Jet Pump vs. Nitrogen Lift – Operational Comparison

Parameter	Jet Pump System (T46S)	Nitrogen Lift (Typical)
Rig-Up Time	1 day	2–3 days
Flow Initiation	Immediate	Delayed, dependent on N_2 injection
Offloading Duration	12 hours	1–3 days
Well Intervention	slickline only for installation	Coiled Tubing + N_2 Units
Operational Cost	Moderate	High
Production Data Quality	High (steady state)	Unstable, transient flow
Environmental and HSE Risk	Low	Moderate to High (N_2 handling)

Flexibility and Adaptability

A key factor in the success of the Well T46S operation was the inherent flexibility of the hydraulic jet pump system. As fluid dynamics shifted from high water cut to oil-dominant flow, the system responded effectively without requiring any mechanical changes or downhole intervention. Surface-based tuning, through adjustments in injection pressure and pump rate, proved sufficient to maintain the desired drawdown and stable production. This level of operational adaptability is especially valuable during early production or cleanup phases, where flow profiles are often unstable and unpredictable (Pugh, 2009; Shen, 2010).

The ability of the jet pump to handle changing fluid conditions without impacting system performance or hardware longevity added significant value to the operation. This is consistent with observations in similar SPE-documented deployments where jet pump reliability under varying flow regimes has enabled extended runtime without intervention (Chavan et al., 2012; Kalwar et al., 2022).

Several technical learnings emerged from this case that can serve as reference for future applications:

- Hydraulic jet pumps, when properly modeled and configured, can be highly effective for early-stage unloading and production testing. Their rigless deployment, compact surface footprint, and non-intrusive retrieval via slickline make them ideal in cost-sensitive or infrastructure-constrained environments (Clark et al., 2014; Shen et al., 2010).
- The accuracy of pre-job simulation using JEMS was critical in optimizing nozzle/throat selection and determining initial surface injection conditions. These models provided a strong foundation for field deployment, supporting both pump efficiency and production forecasting.
- Real-time surface monitoring, including pressure, flow rate, and separator output, enabled operators to quickly detect and address operational anomalies such as backpressure buildup or signs of cavitation. This capability was instrumental in preventing performance degradation and ensuring continuous, optimized drawdown (Mei et al., 2017; Takács, 2015).
- Finally, the case underscored the importance of integrated planning, involving close coordination between field personnel, reservoir engineers, and jet pump specialists to align expectations, validate assumptions, and execute with precision.

In summary, the operational success of this deployment highlights the strategic advantages of jet pump systems in unpredictable flow environments, where responsiveness, simplicity, and surface control are critical to maximizing early production potential and minimizing intervention-related costs.

Potential Challenges in Jet Pump Installation

While the jet pump application in Well T46S was executed successfully, it is important to recognize several technical and operational challenges that can arise during installation and operation. Acknowledging these risks is critical for future planning, prevention strategies, and maintaining consistent system performance across varying well environments.

One of the primary operational challenges in jet pump systems is the requirement for clean, debris-free power fluid, particularly during initial startup. Contaminants such as solids, scale particles, or residual drilling debris can enter the nozzle or throat, leading to partial or full blockage. Such restrictions can significantly reduce pump efficiency, introduce turbulence, or cause complete system failure. To mitigate this, a robust filtration and fluid management protocol must be implemented, including regular maintenance of filtration units and sampling for solids content during injection (Mei et al., 2017; Chavan et al., 2012).

Another common issue relates to accurate pump seating within the downhole Sliding Sleeve Door (SSD). Improper seating can lead to poor hydraulic sealing, inefficient flow paths, or loss of drawdown capability. This is especially critical in older wells, where SSD wear or corrosion may affect the internal geometry. Pre-deployment function testing, visual inspection, and precise slickline depth control are essential to avoid misalignment or loss of pressure integrity (Shen, 2010; Shen et al., 2010).

Power fluid pressure management represents another operational sensitivity. Excessive injection pressure may place stress on tubing, cause surface system failures, or compromise packer integrity, particularly in high-rate or shallow wells. On the other hand, insufficient pressure may result in inadequate drawdown, leading to cavitation and unstable flow behavior. Surface systems must therefore include real-time pressure monitoring, pressure relief valves, and automated flow controls to ensure safe operation within design envelopes (Takács, 2015; Clark et al., 2014).

In some applications, thermal effects and fluid compatibility can also pose challenges. Injecting a power fluid with different temperature, salinity, or chemical composition from the formation fluids may induce scaling, emulsification, or paraffin deposition, compromising flow efficiency. Long-term use of produced water as a power fluid can also lead to erosion of internal pump components, particularly under high solids loading or with abrasive fines. Proper chemical treatment, material selection, and solids monitoring can help manage these risks (Pugh, 2009).

Finally, while slickline deployment and retrieval offer significant cost and safety advantages, they also introduce mechanical constraints. Deep or highly deviated wells may experience retrieval force limitations, or difficulties in pump recovery due to string hang-up or flow path restrictions. In such cases, tool design for optimal slickline conveyance, contingency planning for fishing, and careful evaluation of the well profile are necessary to avoid tool sticking or incomplete retrieval (Kalwar et al., 2022).

Despite these potential challenges, most risks are manageable through a combination of pre-job planning, simulation validation, QA/QC protocols, and live performance monitoring. In the case of Well T46S, proactive implementation of these practices contributed directly to the safe, efficient, and uninterrupted performance of the jet pump system. Table 4 outlines the challenges that may be faced during jet pump operation and their mitigation strategy.

Table 4—Jet Pump Installation – Challenges and Risk Mitigation Measures

Challenge	Description	Mitigation Strategy
Power Fluid Contamination	Solids, scale, or debris in power fluid may plug nozzle/throat.	Use inline filtration; maintain clean tanks; conduct regular fluid sampling.
Improper Pump Seating in SSD	Misalignment or poor engagement may reduce flow efficiency or cause failure.	Perform dry-run testing at surface; verify SSD profile and condition before deployment.
Over- or Under-Pressuring the System	Incorrect surface injection pressure can cause tubing stress or cavitation.	Design surface systems with pressure control valves and monitoring; simulate scenarios.
Erosion from High Solids Content	Continuous flow of abrasive particles may erode internal components.	Monitor solids content; switch to cleaner power fluid or reduce cycle time.
Incompatibility of Injected Fluids	Temperature or chemical mismatches can lead to scaling or emulsion formation.	Match power fluid chemistry to produced fluid; conduct lab compatibility tests.
Slickline Retrieval Limitations	Risk of stuck tool or retrieval failure in deviated/deep wells.	Confirm well trajectory and pull limits; use centralizers or overshot retrieval tools.
Surface Equipment Malfunction	Pump failure, leaks, or control system errors can halt operations.	Conduct thorough function testing; maintain backup equipment and spares.

Conclusion

The deployment of a hydraulic jet pump in Well T46S proved to be a cost-effective, technically robust, and operationally efficient solution for well offloading and early production testing in southern Iraq. In contrast to traditional nitrogen lift using coiled tubing, the jet pump system provided immediate flowback, eliminated the need for extended rig time, and significantly reduced operational costs and logistical complexity.

The well was successfully offloaded within 12 hours, and production stabilized at 2,200 BPD, more than double the initial design target. In parallel, the water cut declined from 80% to 13%, confirming effective drawdown, cleanup, and reservoir inflow activation. All system optimization was conducted at surface, with no need for retrieval or downhole intervention, underscoring the system's modularity, safety, and adaptability (Pugh, 2009; Kalwar et al., 2022).

The system also demonstrated resilience to changing production conditions, including multiphase flow transition and variable backpressure. Key challenges typically associated with jet pump applications, such as solids handling, cavitation control, and fluid compatibility, were effectively managed through a combination of pre-job simulation using JEMS, proactive surface diagnostics, and real-time performance monitoring (Chavan et al., 2012; Mei et al., 2017).

This field case confirms that hydraulic jet pumps are a viable and scalable artificial lift solution, particularly well suited for operations where rigless intervention, fast deployment, and early production diagnostics are critical. The operational and economic success observed in Well T46S supports the broader adoption of jet pump systems not only for short-term well unloading but also for extended testing and long-term artificial lift strategies in similar reservoirs.

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Appendix

Form of Lift	Plunger-Lift System	Capillary Injection System	Electric Submersible Pumping	Reciprocating Rod Pump	Progressing Cavity Pumping	Hydraulic Reciprocating	Jet-Pump System	Gas-Lift System
Maximum operating depth, TVD (ft/m)	19,000 5,791	22,000 6,705	15,000 4,572	16,000 4,878	7,500 2,286	17,000 5,182	15,000 4,572	18,000 4,572
Maximum operating volume (BFPD)	200	500	60,000	6,000	4,500	8,000	35,000	50,000
Maximum operating temperature (°F/°C)	550° 288°	400° 204°	400° 204°	550° 288°	250° 121°	550° 288°	550° 288°	450° 232°
Corrosion handling	Excellent	Excellent	Good	Good to excellent	Fair	Good	Excellent	Good to excellent
Gas handling	Excellent	Excellent	Fair	Fair to good	Good	Fair	Good	Excellent
Solids handling	Fair	Good	Fair	Fair to good	Excellent	Fair	Good	Good
Fluid gravity (°API)	>15°	>8°	>10°	>8°	<40°	>8°	>8°	>15°
Servicing	Wellhead catcher or wireline	Capillary unit	Workover or pulling rig	Workover or pulling rig	Workover or pulling rig	Hydraulic or wireline	Hydraulic or wireline	Wireline or workover rig
Prime mover	Well's natural energy	Well's natural energy	Electric motor	Gas or electric	Gas or electric	Multicylindar or electric	Multicylindar or electric	Compressor
Offshore application	N/A	Good	Excellent	Limited	Limited	Good	Excellent	Excellent
System efficiency	N/A	N/A	35% to 60%	45% to 60%	50% to 75%	45% to 55%	10% to 30%	10% to 30%

Note: These results represent a only generated simulation, which should not be taken as actual advice for your wells. Please contact Weatherford for customized, application-specific expertise.

Figure A—Limitations and capabilities of artificial lift systems

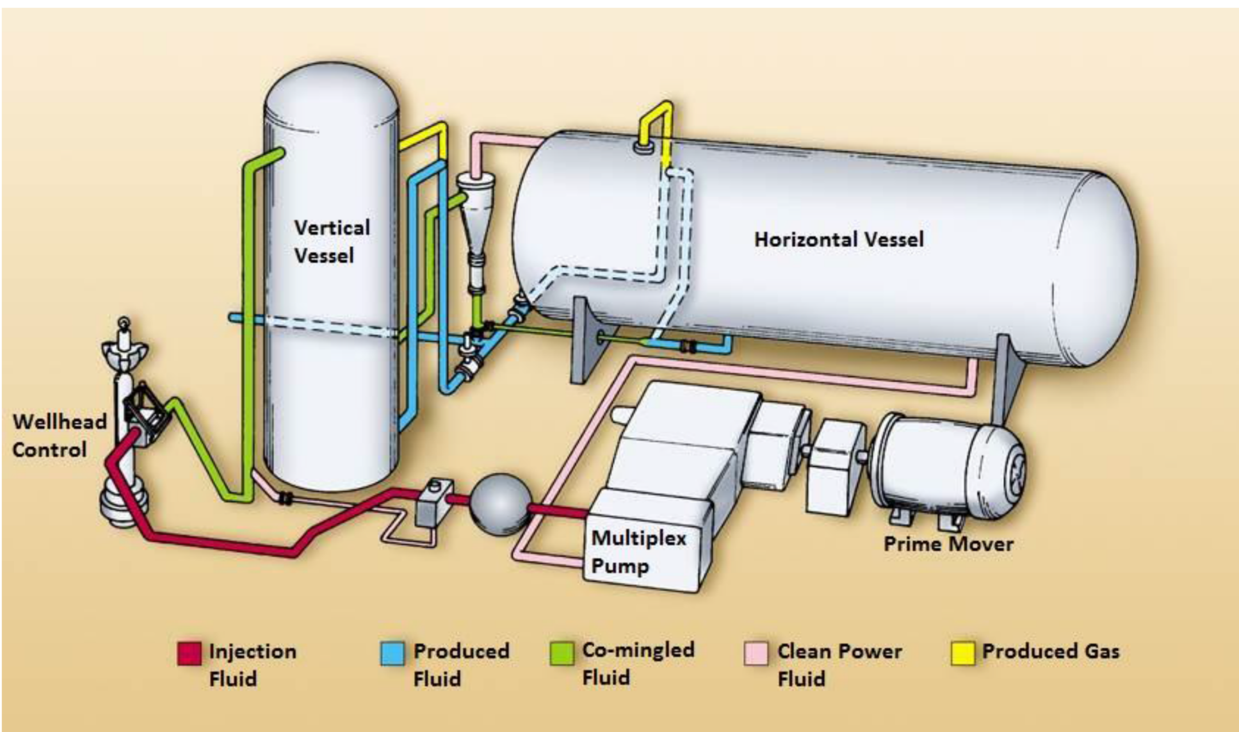

Figure B—Surface Setup Schematic for Jet Pump System

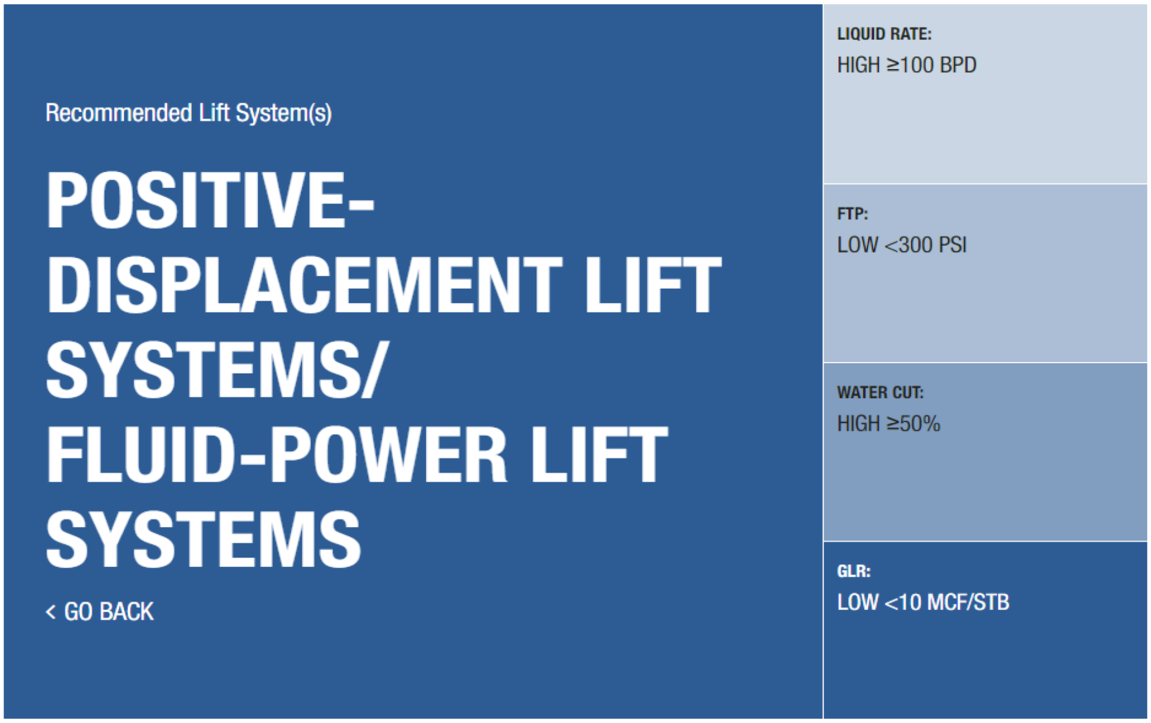

Figure C—Weatherford Unloading Selector

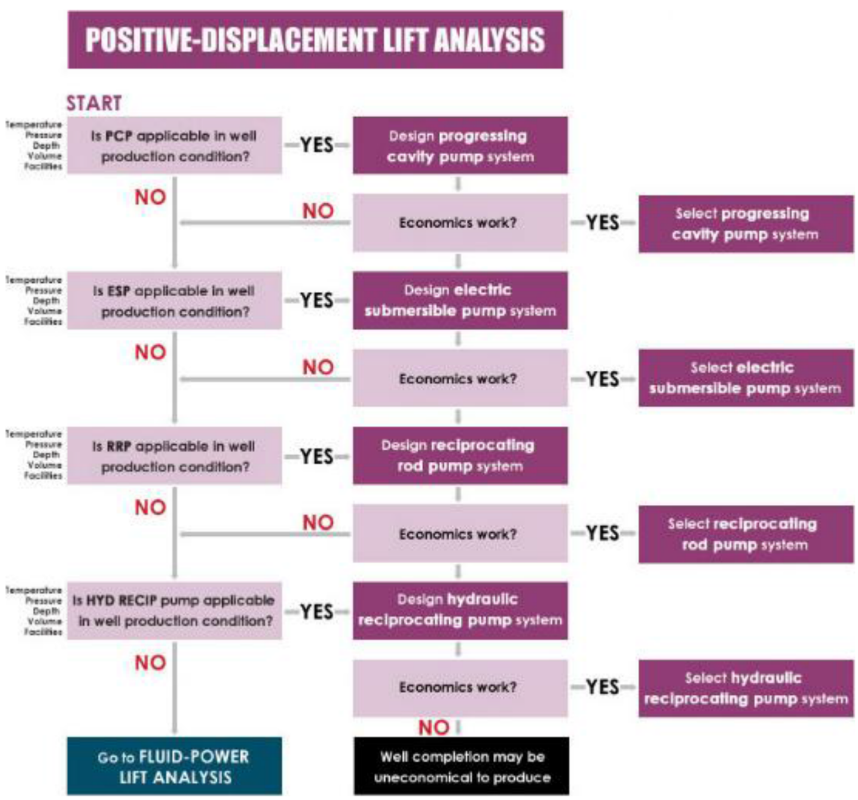

Figure D—Positive Displacement Lift Analysis

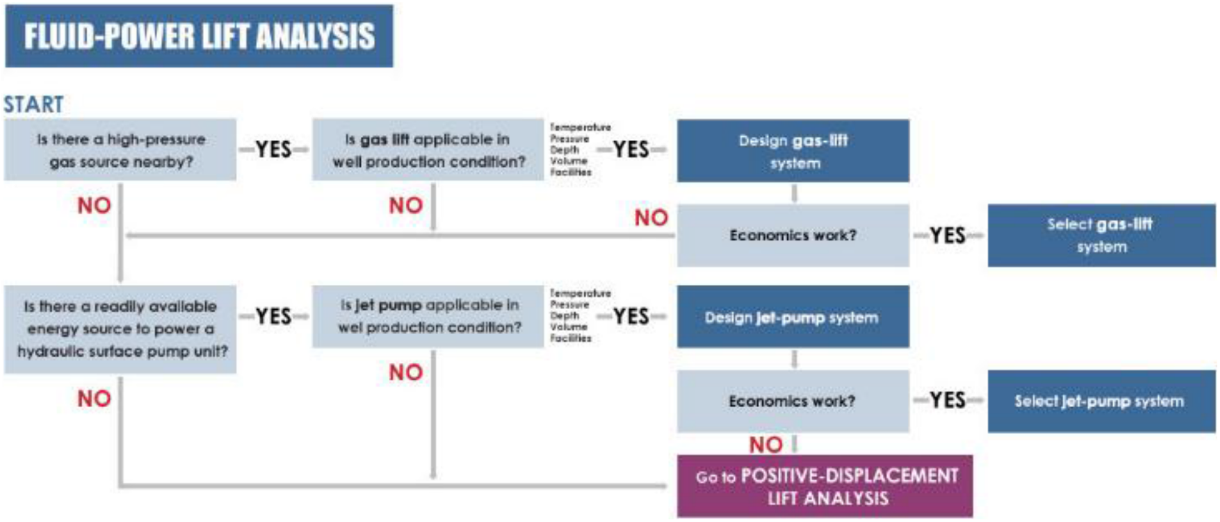

Figure E—Fluid Power Lift Analysis